Discrete Drainage Dynamics: A Bounded, Conservative Local Rule for Newtonian-Like Potentials

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Abstract

We define a deterministic, parallel, local update rule on a discrete graph. Each node carries a non-negative reserve $R_i \geq 0$, with rest value R_0 at equilibrium. Each oriented link carries a non-negative directional flux. The rule alternates a node update and a link update, with the outgoing fluxes from a node rate-limited so that the reserve cannot become negative. The flux on a link is written in a general form involving a *mobility function* $\mu(R)$ which encodes how the source node's reserve modulates its emission rate. The present paper adopts the constant-mobility baseline $\mu \equiv 1$, the simplest choice consistent with the substrate's locality and conservation properties. In this baseline, far from the non-negativity floor, the rule's steady state coincides with the discrete Poisson equation, and we measure on the lattice that the static profile around a point source follows a $1/r$ law with the continuum coefficient at the percent level. The contribution of this paper is the rule and the mobility framing, not the $1/r$ law. Companion papers study other mobility prescriptions and their non-linear consequences.

1 What this paper claims

We make three claims.

1. We define a deterministic, parallel, two-step local rule on a discrete graph. The reserve is bounded below by zero ($R_i \geq 0$, the physical floor: one cannot drain what is not there); R_0 is the rest value, but the reserve is free to exceed it in excited regimes treated in companion papers. The flux on a link involves a mobility function $\mu(R)$ that we leave as a structural choice of the framework; the present paper studies the constant-mobility baseline $\mu \equiv 1$.

2. Run as a stationary linearised simulation with a point source on a cubic lattice, the rule (with $\mu \equiv 1$) produces a radial profile compatible with $1/r$ in the bulk, with a coefficient that matches the discrete continuum prediction at the percent level across $L \in \{64, 128, 160\}$.

3. The internal flux-exchange rule preserves total reserve exactly. Non-negativity of the reserve is enforced by rate-limiting the outgoing fluxes and external sinks of a node in proportion, so that emission stops when the reserve reaches zero; no mass is created or destroyed by the internal rule. This is the structural feature that distinguishes the rule from a plain discrete heat equation.

We do not claim the following. We do not derive Newton's gravitational constant in physical units; we calibrate it. We do not derive general relativity. We do not derive any property of the Standard Model. We do not show that the rule is unique. We do not derive the mobility function $\mu(R)$; the constant-mobility baseline of this paper is a choice within a family of possibilities. We do not assert that the $1/r$ result is itself novel: it is not. What is novel is the rule, the mobility framing, and the structural features (reserve non-negativity, rest level R_0, two-step alternation, rate-limited fluxes) that make the rule non-trivial in the regimes explored in the companion papers.

Motivation and significance

The purpose of this paper is *not* to rediscover the $1/r$ potential. Any linear diffusion or graph-Laplacian model in three dimensions has a Green function with this behaviour. The purpose is instead to define a local, conservative substrate rule whose weak-field limit reduces to the discrete Poisson equation, while retaining *finite-resource* structure outside that limit.

This distinction matters for the rest of the DDD programme. A plain heat equation can reproduce $1/r$, but it has no physical floor, no finite local budget, no source-side mobility, and no natural saturation regime. The rule introduced here adds these ingredients: every node carries a non-negative reserve $R \geq 0$, every outgoing flux is rate-limited by the available local budget, and the mobility $\mu(R)$ controls how the source node's reserve modulates emission. These are the substrate features on which the companion papers depend.

The baseline $\mu \equiv 1$ studied in this paper is therefore a *calibration and validation regime*. It verifies that the rule reduces to the expected Poisson behaviour when the finite-resource constraint is inactive. The non-trivial physics begins when $\mu(R) \neq 1$ (Paper II's clock-rate suppression and Schwarzschild geometry), or when the reserve floor $R = 0$ becomes active (saturation, horizon-like bandwidth limit), or when the spinor and gauge-sector back-reactions of Papers III, VI, and VII activate. Paper I supplies the conservative local substrate on which they all depend.

A diagnostic of this finite-resource structure is that superposition of source profiles, exact in the linear regime, must *break down locally* when the reserve floor or the mobility nonlinearity becomes active. We exhibit this directly in §6.4 as a sanity check.

2 The picture, in plain words

Imagine a fine three-dimensional grid. At each grid point sits a reservoir holding a non-negative amount of stuff. There is a rest level R_0 that the reservoirs all sit at when the system is undisturbed; reservoirs can be excited above R_0 or drained below it, but they cannot go below zero. Between neighbours run pipes that carry a non-negative flow of stuff from the fuller reservoir to the emptier one, at a rate proportional to the level difference, optionally modulated by how much stuff the source actually holds. A reservoir cannot deliver more stuff than it actually has: a node at zero stops emitting.

If we start the system at the rest level R_0 everywhere, drain one point, and let the system settle, the bulk reservoir levels relax into a stationary pattern. With the simplest choice of mobility ($\mu \equiv 1$, the case of this paper), far from the source the pattern reduces to the textbook discrete Poisson solution and the deficit $R_0 - R_i$ falls as $1/r$ with distance. Other choices of mobility yield qualitatively different non-linear regimes, studied in companion papers.

The interesting cases are where the floor matters. Near a strong drainage point, the reservoir level can be driven to zero, and emission from that node stops. In this regime the rule is no longer a discrete heat equation: it is a non-linear update with a finite-resource bound. The companion papers explore those regimes (Schwarzschild-form behaviour in Paper II, Floquet spinor sector in Paper III, gauge sector in Paper VI, gravity-gauge coupling in Paper VII). The present paper is restricted to the linearised regime with $\mu \equiv 1$, where the rule reproduces $1/r$.

3 The substrate

3.1 Graph and variables

The substrate is a graph $G = (V, E)$ in which every node has at least six neighbours and the connectivity is statistically isotropic at large scale. We do not specify the exact topology; in our simulations we use a regular cubic lattice for clarity, and we have verified that BCC gives the same large-scale spectral dimension.

For the purposes of this paper, each node $i \in V$ carries a non-negative reserve $R_i \geq 0$, with rest value R_0 at equilibrium. Each oriented link $(i, j) \in E$ carries a pair of non-negative directional reserve fluxes $F_{i\to j} \geq 0$ and $F_{j\to i} \geq 0$. These are the only variables that enter the linearised gravitational simulation reported here. The non-negativity $R_i \geq 0$ is the physical floor (one cannot drain what is not there).

Variables of the wider rule

The full rule, as used in the companion papers, also assigns to each node two angular quadratures (T_i, I_i) subject to a local coherence bound $T_i^2 + I_i^2 \leq R_i$, and to each oriented link a phase $\psi_{ij} \in [-\pi, \pi)$ with $\psi_{ji} = -\psi_{ij}$. These additional variables drive the gauge sector (Papers V–VI), couple to the reserve through the coherence bound, and do not enter the static linearised gravitational regime treated below.

3.2 Parameters

Parameter	Role	Used in this paper
κ	drainage coefficient	yes
α	flux propagation coefficient	yes
R_0	reserve rest value (equilibrium)	yes
$\mu(R)$	mobility function (structural choice)	yes ($\mu \equiv 1$)
J	link relaxation coupling	no (gauge sector, deferred)

Table 1: Parameters of the local rule. The reserve floor at 0 is not a parameter: it is the physical statement that one cannot drain what is not there. R_0 is the rest level at equilibrium, not an absolute ceiling: the reserve can exceed R_0 in excited regimes treated in the companion papers. The mobility $\mu(R)$ is a structural choice of the framework: this paper uses $\mu \equiv 1$, companion papers (notably Paper II) use other choices.

4 The two-step tick

Motivation

The class of rules considered here is motivated by an old line of inquiry on whether physics admits a microscopic deterministic, local, iterative description — a "computational substrate" in the sense of Zuse, Fredkin, 't Hooft, Wolfram, and Lloyd [?, ?, ?, ?, ?]. We do not commit to the strong form of this hypothesis. We adopt it as a working stance.

The rule

The rule decomposes into two alternating half-steps, which we call *Node Update* and *Link Update*. The choice of two-step alternation is a standard one for local lattice theories: it preserves locality and avoids self-reference within a tick.

4.1 Link Update

During the first half-tick, the node reserves are frozen. For each oriented link (i, j), in parallel, we compute a desired flux. We write the desired flux in the general form

$$F^{\text{des}}_{i\to j} \;=\; \alpha\,\mu(R_i)\,(R_i - R_j)_+, \tag{1}$$

where $\mu(R)$ is a non-negative *mobility function* encoding how the available reserve at the source node limits its ability to push reserve into a neighbour. We normalise $\mu(R_0) = 1$ at the rest level, so that α retains its meaning as the rated propagation coefficient when the source is at rest. We do not derive $\mu(R)$ from a deeper principle; it is a structural choice of the rule.

Constant-mobility baseline of this paper. The present paper adopts the constant-mobility baseline,

$$\mu(R) \;\equiv\; 1, \tag{2}$$

for which Eq. (1) reduces to the textbook discrete-diffusion form $F^{\text{des}}_{i\to j} = \alpha(R_i - R_j)_+$, and the linearised steady state is the discrete Poisson equation, giving the $1/r$ profile measured in Sec. 6.

Other mobility choices in companion papers. The reserve-proportional mobility $\mu(R) = R/R_0$, motivated by local bandwidth limitation at the source node, is studied in Paper II, where it is shown to produce $\nabla^2(R^2) = 0$ in the continuum limit and a profile of Schwarzschild form. Other mobility prescriptions may be explored in subsequent papers. Within Paper I, none of these is invoked.

The two directions of a link are independent. Each node then computes a rate-limiting factor that ensures the reserve stays non-negative on its outgoing side. Define

$$A_i \;=\; R_i \,+\, \tau\, S^{\text{in}}_i, \quad S^{\text{in}}_i = \sum_j F^{\text{des}}_{j\to i}, \quad D_i \;=\; \tau\, O^{\text{des}}_i + \tau\,\kappa\, E_i, \quad O^{\text{des}}_i = \sum_j F^{\text{des}}_{i\to j}. \tag{3}$$

The rate-limiter is

$$\beta_i \;=\; \begin{cases} 1, & D_i = 0, \\ \min(1,\; A_i/D_i)\,, & D_i > 0, \end{cases} \tag{4}$$

applied identically to outflows and sink:

$$F_{i\to j} \;=\; \beta_i\, F^{\text{des}}_{i\to j}, \qquad E^{\text{real}}_i \;=\; \beta_i\, E_i. \tag{5}$$

4.2 Node Update

During the second half-tick, the link fluxes are frozen.

$$R_i \;\leftarrow\; R_i + \tau\, S^{\text{in}}_i \;-\; \beta_i\,\tau\, O^{\text{des}}_i \;-\; \beta_i\,\tau\,\kappa\, E_i. \tag{6}$$

By construction of β_i, this update keeps R_i non-negative at every tick.

5 Conservation

Total reserve, exactly conserved by internal exchange. The realised fluxes enter the Node Update of j as $+\beta_i F^{\text{des}}_{i\to j}$ and the Node Update of i as $-\beta_i F^{\text{des}}_{i\to j}$. Summed over the graph, $\sum_i R_i$ is exactly conserved by inter-node flux dynamics. The rate-limiter scales outflows and the local sink in the same proportion.

External sinks as bookkeeping. The total reserve in the network changes only by $\sum_i E^{\text{real}}_i$. The closed system network-plus-external-sink is conservative; the network alone is open.

6 Simulation: from rule to $1/r$

6.1 Derivation of the Poisson limit (constant mobility)

We work in the constant-mobility baseline $\mu \equiv 1$. In the linear regime $\beta_i = 1$, the two directional fluxes satisfy $F_{i \to j} - F_{j \to i} = \alpha(R_i - R_j)$, so $S_i = \alpha \Delta R_i$ where Δ is the discrete graph Laplacian. In stationary regime, $\alpha \Delta R_i = \kappa E_i$. Switching to the deficit $\delta_i = R_0 - R_i$:

$$-\Delta\, \delta_i \;=\; M\, \delta_{\mathrm{Kr}}(i, i_0), \qquad M \;=\; \frac{\kappa E_0}{\alpha}. \tag{7}$$

The continuum solution is $\delta(r) = M/(4\pi r)$.

6.2 Numerical check

L	A_{fit}	relative bias	statistical precision	N_{iter}
64	0.07878	-1.01%	7.0×10^{-5}	5 000
128	0.07866	-1.16%	5.7×10^{-6}	10 000
160	0.07886	-0.90%	3.7×10^{-6}	30 000
$A_{\mathrm{continuum}} = M/(4\pi)$		0.07958		

Table 2: Newton-fit coefficients on cubic lattices, $M = 1$, Dirichlet boundary, with $\mu \equiv 1$. The bias is roughly L-independent at the percent level (Madelung correction).

6.3 Robustness: spectral dimension

L	cubic ($d_s - 3$ minimum)	BCC ($d_s - 3$ minimum)
32	0.129	0.107
64	0.054	0.045
128	0.023	0.019

Table 3: Spectral-dimension residual converges to 0 in the IR limit on both substrates. Newtonian gravity recovered.

6.4 Two-source superposition and onset of finite-resource effects

As a second sanity check that distinguishes the DDD rule from a plain graph-Laplacian, we place two identical point sinks at separation $d = 8$ lattice spacings and compare the steady-state deficit field δ_{12} with the linear superposition $\delta_1 + \delta_2$ obtained from two independent one-source runs.

Linear regime ($\mu = 1$, weak sinks). For sink strength 0.001, the reserve floor is never approached ($R_{\min} \simeq R_0 - \mathcal{O}(10^{-4})$, with $R_{\min}/R_0 > 0.999$ everywhere), the rate-limiter $\beta_i \equiv 1$ throughout, and the rule reduces to the discrete Poisson equation. The relative residual $\max|\delta_{12} - (\delta_1 + \delta_2)|/\max \delta_{12}$ is 5.4×10^{-3}, attributable entirely to lattice and boundary truncation. Superposition holds.

Saturating regime (rate-limiter active). For sink strength 12, the reserve at each source is driven down to the floor $R = 0$. The rate-limiter $\beta_i = \min(1, A_i/D_i)$ (where A_i is the available reserve and D_i is the demand from outgoing fluxes plus external sinks at node i) drops below 1 at depleted nodes, capping the outgoing flux, and the steady-state field no longer obeys superposition: the relative residual rises to 2.7×10^{-2}, an order of magnitude above the lattice-error baseline, and is concentrated near the saturated sources where δ_1 and δ_2 would otherwise have added linearly.

This contrast — $\sim 0.5\%$ residual when the floor is inactive vs $\sim 3\%$ when it is active — is the diagnostic signature of finite-resource physics that motivates the rule. The deviation is not a numerical defect; it is the substrate correctly refusing to deliver more flux than its local reserve permits. Figure 1 displays the central slice of both regimes side-by-side.

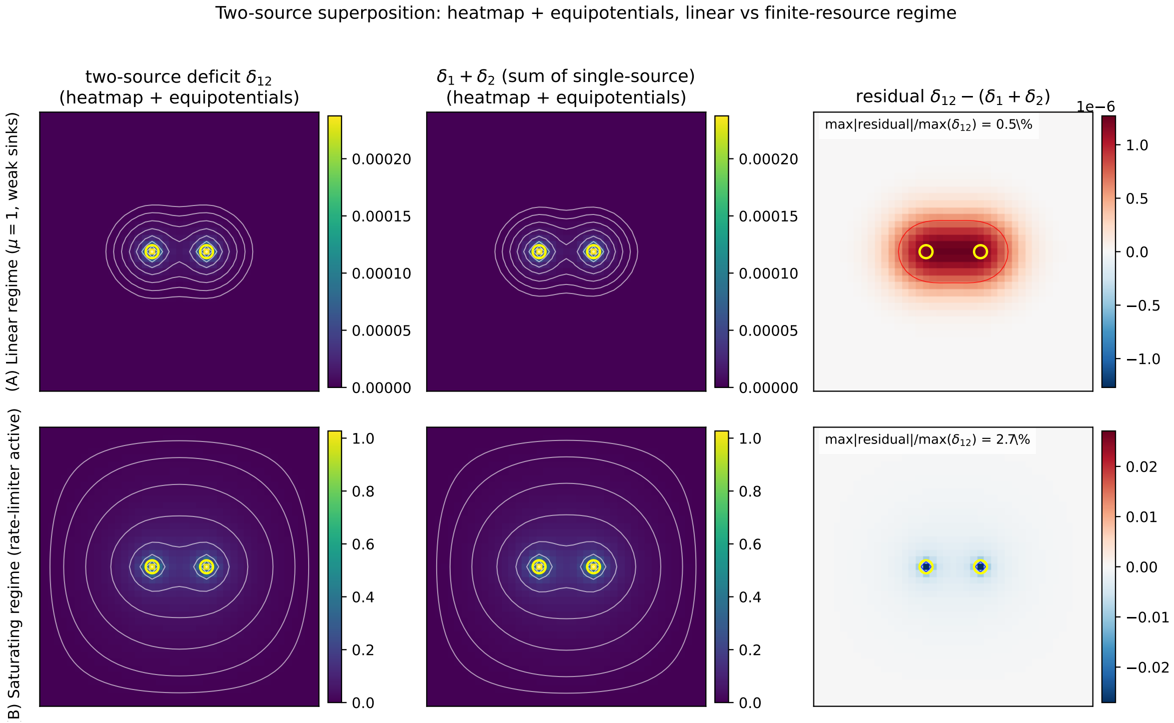

Figure 1: Two-source superposition test. **Top row (A)**, linear regime ($\mu = 1$, weak sinks): the two-source deficit δ_{12} (left), the superposition $\delta_1+\delta_2$ (centre), and the residual (right). The residual is at the 0.5% level and consists entirely of lattice/boundary error. **Bottom row (B)**, saturating regime (rate-limiter active): the residual rises to 2.7% and is localised at the sources where the reserve floor caps the outgoing flux. *Heatmap + equipotentials*: white contour lines mark log-spaced equipotentials of the deficit field; on the residual panel, the black contour is the zero line and the red/blue contours mark $\pm 50\%$ of the residual maximum. Source positions marked with yellow circles. Plane shown is $z = z_{\text{centre}}$. Colourmaps: viridis for absolute deficit, RdBu for the residual (symmetric about zero). Reproducible via `code/11_two_source_superposition.py`.

6.5 Code and data availability

Scripts available at `https://github.com/stanislasdewavrin/DDD-Paper-1-Foundations`. Pure-numpy, no scipy.

7 Calibration vs. prediction

Calibration. The mappings $\delta \leftrightarrow \Phi$ and $E \leftrightarrow m$ are external. The lattice unit and κ are fixed by demanding A_{fit} matches Newton's G.

Prediction. Within that family, the rule (with $\mu \equiv 1$) predicts that the static potential is $1/r$ on the linearised regime.

8 What the model is not, and what it does not do

We do not derive general relativity, Newton's G, the speed of light c, or any property of the Standard Model.

We do not specify the exact graph topology beyond "minimum 6 neighbours, statistically isotropic". We have tested cubic and BCC.

We do not derive the mobility function $\mu(R)$. The present paper studies the constant-mobility baseline $\mu \equiv 1$, for which the linearised steady state is the discrete Poisson equation. Different mobility choices lead to different non-linear steady-state equations; the constant-mobility choice is the minimal Newtonian baseline. Companion papers study specific reserve-dependent mobility prescriptions, in particular the source-side bandwidth-throttled choice $\mu(R) = R/R_0$ in Paper II.

We have not shown that the rule is unique.

9 Open questions

Why six neighbours? Six is the minimal cubic coordination; not a fundamental requirement. More general graphs require separate checks of spectral dimension and isotropy.

Why this order, Node then Link? Microscopic breaking of time-reversal symmetry; we have no derivation.

What sets R_0? The only structural scale of the rule. Not derived.

What sets the mobility $\mu(R)$? The constant-mobility baseline $\mu \equiv 1$ used here is the simplest choice. Other choices (in particular the source-side bandwidth-throttled $\mu(R) = R/R_0$ studied in Paper II) yield qualitatively different non-linear regimes. A microscopic argument that selects a specific mobility from the rule would close an important gap of the framework.

Is the rule unique? See Sec. 8.

What sits below the substrate? Beyond the present paper.

10 Outlook

1. **Paper II** — the source-side reserve-proportional mobility $\mu(R) = R/R_0$ yields a Schwarzschild-like static clock-rate factor in the spherical continuum limit, recovering weak-field general-relativistic time dilation from the local substrate rule. *Conditional on the mobility choice and on the bandwidth-time identification.*

2. **Paper III** — a Floquet two-step spinor sector on the same substrate supports propagating localised excitations and an operational, kinematic Lorentz-like clock-rate.

3. **Paper IV** — calibrates the photon-substrate coupling against the static spherical sector and reproduces light deflection and Shapiro delay at the percent level.

4. **Paper V** — collects candidate predictions and falsification windows of the gravitational sector (Yukawa bounds, short-range tests, feedback-mediated effects).

5. **Paper VI** — extends the substrate with a compact U(1) oriented sector, derives Maxwell-like dynamics in the weak-field limit, and proposes a candidate fixed-point formula for the fine-structure constant $\alpha_{\mathrm{EM}} \simeq 1/137$ based on the body-diagonal Wilson loop self-energy.

6. **Paper VII** — studies the joint Wilson-loop back-reaction of gravity and electromagnetism at the substrate scale, predicting a structural ratio $\alpha_G/\alpha_{\mathrm{EM}} = 1/\sqrt{2}$ and cross-coupling consequences observable in atomic clocks and cosmological $\dot{\alpha}/\alpha$.

The present paper, Paper I, supplies only the bounded conservative local rule and its constant-mobility Poisson baseline, validated by the $1/r$ verification of §6 and by the two-source superposition sanity check of §6.4. The support level and speculative content vary across the later papers and are stated explicitly in each.